Amelia A. Miramonti

Curriculum Vitae | Postdoctoral Researcher and Adjunct Faculty

Nebraska Evaluation and Research (NEAR) Center · University of Nebraska–Lincoln
University of Nebraska Public Policy Center (NUPPC)
Nebraska Statewide Workforce and Educational Reporting System (NSWERS)
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Education

PhD, Interdisciplinary - University of Nebraska–Lincoln- Lincoln, NE — 2015–2025

- Majors: Quantitative, Qualitative, & Psychometric Methods (Educational Psychology)
- Human Sciences: Exercise Physiology & Nutrition (Nutrition & Health Sciences)
- Dissertation: Youth Sport Participation and Cognitive Development: The Need for Improved Measurement
- Conferred May 17, 2025

MA, Quantitative, Qualitative, and Psychometric Methods - University of Nebraska–Lincoln- Lincoln, NE — 2018

- Educational Psychology
- Thesis: Evaluation of the Feasibility of a Two-Method Measurement Design for the Assessment of Healthy Physical Activity Behavior in Youth

MSc, Sport & Exercise Science - University of Central Florida- Orlando, FL — 2013–2015

- Applied Exercise Physiology Track
- Thesis: The Effects of Four Weeks of High-Intensity Interval Training and β-Hydroxy-β-Methylbutyric Free Acid Supplementation on the Onset of Neuromuscular Fatigue

BSc, Sport and Exercise Science - University of Central Florida- Orlando, FL — 2007–2013

- Human Performance Specialization

Academic Professional Experience

Adjunct Faculty Instructor - Department of Educational Psychology- University of Nebraska–Lincoln — January 2026–present

- EDPS 900D – Survey Methods in Educational Research (Fall 2026); EDPS 845 – Computer-Assisted Research Data Analysis (Spring 2026).

Postdoctoral Researcher - Nebraska Evaluation and Research Center (NEAR Center), Department of Educational Psychology- University of Nebraska–Lincoln — July 2025–present

- Cross-disciplinary appointment integrating research design and methodology, statistics and data science, and public policy. Primary responsibilities include collaborating on funded research projects, contributing to research design, data analysis, and reporting, and supervising graduate research assistants (currently five) in the Quantitative, Qualitative, and Psychometric Methods (QQPM) program.

Postdoctoral Research Associate (contracted affiliation) - Nebraska Statewide Workforce & Educational Reporting System (NSWERS)- University of Nebraska–Lincoln — July 2025–present

- Creating privacy-preserving synthetic datasets from large-scale, longitudinal administrative data to support public research initiatives.

Postdoctoral Research Associate (contracted affiliation) - University of Nebraska Public Policy Center (NUPPC)- University of Nebraska–Lincoln — July 2025–present

- Analysis, data management, and reporting across Center projects serving Nebraska state agencies and communities: Nebraska Suicide Mortality Review; Ryan White Clinical Quality Management program; Lincoln Food & Recyclable Waste initiative; Community Crops classes evaluation; and survey design for the Get in the Game project with Lincoln Public Schools Community Learning Centers.

Research Coordinator (full-time) - Nebraska Evaluation and Research Center, Department of Educational Psychology- University of Nebraska–Lincoln — September 2024–July 2025

- Methodological and analytic support across concurrent projects spanning multiple content areas, including graduate thesis and dissertation research and funded studies. Supervised four graduate research assistants, training them in analysis and managing their assignments across projects.

Graduate Research Assistant - Nebraska Evaluation and Research Center, Department of Educational Psychology- University of Nebraska–Lincoln — 2021–August 2024

Graduate Research Assistant - Nebraska Academy for Methodology, Analytics & Psychometrics, Nebraska Center for Research on Children, Youth, Families, & Schools- University of Nebraska–Lincoln — 2018–2021

Graduate Research Assistant - Nebraska Extension: 4-H Youth Development, Institute of Agriculture and Natural Resources- University of Nebraska–Lincoln — Spring 2018

Graduate Research Assistant - Nebraska Extension: Nutrition Education Program/SNAP-Ed, Department of Nutrition & Health Sciences- University of Nebraska–Lincoln — Fall 2017

Graduate Research Assistant - Neuromuscular Research & Imaging Laboratory, Department of Nutrition & Health Sciences- University of Nebraska–Lincoln — 2015–2017

- Study administration and data collection for laboratory and field research with participants from age nine through older adulthood: participant recruitment and scheduling, research staff scheduling, and electromyographic, mechanomyographic, and strength data collection and processing.

Graduate Teaching Assistant - Department of Educational Psychology- University of Nebraska–Lincoln — Spring 2022

- EDPS 459 – Statistical Methods.

Graduate Teaching Assistant - Department of Nutrition & Health Sciences- University of Nebraska–Lincoln — 2015–2017

- NUTR 250 – Human Nutrition & Metabolism (Fall 2015, Fall 2016, Spring 2017); NUTR 484 – Physiology of Exercise Laboratory (Spring 2016); NUTR 486 – Exercise Testing & Prescription Laboratory (Spring 2016).

Graduate Research Assistant - Institute of Exercise Physiology & Wellness, Department of Educational & Human Science- University of Central Florida — 2013–2015

- Study administration and data collection for laboratory research: participant recruitment and scheduling, research staff scheduling, and data collection across metabolic, strength, and neuromuscular assessment protocols.

Graduate Teaching Associate - Department of Educational & Human Science- University of Central Florida — 2014–2015

- PEM 2104 Personal Fitness (Summer 2014); PET 3005 Intro to Sport & Exercise Science (Summer 2014, Fall 2014, Spring 2015); PET 4083 Personal Training Methods (Spring 2015).

Research Practicum - Institute of Exercise Physiology & Wellness, Department of Educational & Human Science- University of Central Florida — Fall 2012

Invited & Guest Lectures

Data documentation for Teams (Session 1 of 2) - Data Management Brown Bagger Series, University of Nebraska Public Policy Center, Lincoln, NE. — June 2026

Interaction Contrasts in ANOVA - Guest lecture, EDPS 941 – Intermediate Statistics: Experimental Methods (Dr. Matt Fritz, instructor), Department of Educational Psychology, University of Nebraska–Lincoln, Lincoln, NE. — March 26, 2026

Meeting the Demands of High Intensity Interval Training: Metabolic and Physiological Adaptations to High-Intensity Interval Training - Lecture, 2017 National Strength & Conditioning Association National Conference, symposium sponsored by GNC Nutrition, Inc., Las Vegas, NV. — July 12–15, 2017

Refereed Journal Articles

Citations: 1,850 | h-index: 18 | i10-index: 22, as of 2026-10-12 (via Google Scholar)

Published

1. Kim, S., Kantamneni, N., Ahmad Ridzuan, N. H., & Miramonti, A. A. (2026). Work-family conflict and burnout for mothers of color: Exploring the role of microaggressions and gender role beliefs. *Journal of Career Assessment*, *34*(1), 3–21. https://doi.org/10.1177/10690727241307957
2. George, R. M., Dev, D. A., Miramonti, A. A., Hasnin, S., Hillburn, C., Smith, J., Sisson, S. B., & Tovar, A. (2025). Family style meal service is associated with reduced plate waste in Nebraska family child care homes. *Childhood Obesity*.
3. Jenkins, N. D. M., Miramonti, A. A., Hill, E. C., Smith, C. M., Cochrane-Snyman, K. C., Housh, T. J., & Cramer, J. T. (2021). Mechanomyographic amplitude is sensitive to load-dependent neuromuscular adaptations in response to resistance training. *Journal of Strength & Conditioning Research*, *35*(11), 3265–3269. https://doi.org/10.1519/jsc.0000000000003276
4. McKay, B. D., Miramonti, A. A., Gillen, Z. M., Leutzinger, T. J., Mendez, A. I., Jenkins, N. D. M., & Cramer, J. T. (2020). Normative reference values for high school-aged American football players: Proagility drill and 40-yard dash split times. *Journal of Strength & Conditioning Research*, *34*(4), 1184–1187. https://doi.org/10.1519/jsc.0000000000002930
5. McKay, B. D., Miramonti, A. A., Gillen, Z. M., Leutzinger, T. J., Mendez, A. I., Jenkins, N. D. M., & Cramer, J. T. (2020). Normative reference values for high school-aged American football players. *Journal of Strength & Conditioning Research*, *34*(10), 2849–2856. https://doi.org/10.1519/jsc.0000000000002532
6. Gillen, Z. M., Miramonti, A. A., McKay, B. D., Leutzinger, T. J., & Cramer, J. T. (2018). Test-retest reliability and concurrent validity of athletic performance combine tests in 6–15-year-old male athletes. *Journal of Strength & Conditioning Research*, *32*(10), 2783–2794. https://doi.org/10.1519/jsc.0000000000002498
7. Leutzinger, T. J., Miramonti, A. A., Gillen, Z. M., McKay, B. D., Mendez, A. I., & Cramer, J. T. (2018). Anthropometric and athletic performance combine test results among positions within grade levels of high school-aged American football players. *Journal of Strength & Conditioning Research*, *32*(5), 1288–1296. https://doi.org/10.1519/jsc.0000000000002481
8. Mangine, G. T., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Miramonti, A. A., Wang, R., La Monica, M. B., Fukuda, D. H., Ratamess, N. A., Stout, J. R., & Hoffman, J. R. (2018). Influence of baseline muscle strength and size measures on training adaptations in resistance-trained men. *International Journal of Exercise Science*, *11*(4), 198–213.

9. Gillen, Z. M., Miramonti, A. A., McKay, B. D., Jenkins, N. D. M., Leutzinger, T. J., & Cramer, J. T. (2018). Reliability and sensitivity of the power push-up test for upper-body strength and power in 6–15-year-old male athletes. *Journal of Strength & Conditioning Research*, *32*(1), 83–96. https://doi.org/10.1519/jsc.0000000000002313
10. Mendez, A. I., Miramonti, A. A., Gillen, Z. M., McKay, B. D., Leutzinger, T. J., & Cramer, J. T. (2017). Stature, body mass, and BMI in high school American football players: Appropriate determinants of obesity prevalence? *Journal of Strength & Conditioning Research*. https://doi.org/10.1519/jsc.0000000000002347
11. Mangine, G. T., Hoffman, J. R., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Boone, C. H., Wang, R., Miramonti, A. A., La Monica, M. B., Fukuda, D. H., Witta, E. E., Ratamess, N. A., & Stout, J. R. (2017). Exercise-induced hormone elevations are related to muscle growth. *Journal of Strength & Conditioning Research*, *31*(1), 45–53. https://doi.org/10.1519/jsc.0000000000001491
12. Wang, R., Fukuda, D. H., Stout, J. R., Robinson, E. H., Miramonti, A. A., Fragala, M. S., & Hoffman, J. R. (2017). Tracking changes in the upper boundary of the heavy-intensity exercise domain: End-test power versus respiratory compensation point. *Kinesiology*, *49*(1), 15–21. https://doi.org/10.26582/k.49.1.13
13. Jenkins, N. D. M., Miramonti, A. A., Hill, E. C., Smith, C. M., Cochrane-Snyman, K. C., Housh, T. J., & Cramer, J. T. (2017). Greater neural adaptations following high- vs. Low-load resistance training. *Frontiers in Physiology*, *8*, 331. https://doi.org/10.3389/fphys.2017.00331
14. McKay, B. D., Yeo, N. M., Jenkins, N. D. M., Miramonti, A. A., & Cramer, J. T. (2017). Exertional rhabdomyolysis in a 21-year-old healthy woman: A case report. *Journal of Strength & Conditioning Research*, *31*(5), 1403–1410. https://doi.org/10.1519/jsc.0000000000001824
15. Miramonti, A. A., Jenkins, N. D. M., Oza, P. D., Weir, J. P., & Cramer, J. T. (2017). Mechanomyographic responses during recruitment curves in the soleus muscle. *Muscle & Nerve*, *56*, 107–116. https://doi.org/10.1002/mus.25432
16. Riffe, J. J., Stout, J. R., Fukuda, D. H., Robinson, E. H., Miramonti, A. A., Beyer, K. S., Wang, R., Church, D. D., Muddle, T. W. D., & Hoffman, J. R. (2017). The Dmax method is a valid procedure to estimate physical working capacity at fatigue threshold. *Muscle & Nerve*, *55*(3), 344–349. https://doi.org/10.1002/mus.25255
17. Miramonti, A. A., Stout, J. R., Fukuda, D. H., Robinson, E. H., Wang, R., La Monica, M. B., & Hoffman, J. R. (2016). Effects of 4 weeks of high-intensity interval training and β-hydroxy-β-methylbutyric free acid supplementation on the onset of neuromuscular fatigue. *Journal of Strength & Conditioning Research*, *30*(3), 626–634. https://doi.org/10.1519/jsc.0000000000001140
18. Beyer, K. S., Fukuda, D. H., Miramonti, A. A., Hoffman, M. W., Wang, R., La Monica, M. B., Riffe, J. J., Tanigawa, S., Stout, J. R., & Hoffman, J. R. (2016). Spatial awareness is related to moderate intensity running during a collegiate rugby match. *International Journal of Exercise Science*, *9*(5), 599.
19. Smith, C. M., Housh, T. J., Jenkins, N. D. M., Hill, E. C., Cochrane, K. C., Miramonti, A. A., Schmidt, R. J., & Johnson, G. O. (2016). Combining regression and mean comparisons to identify the time course of changes in neuromuscular responses during the process of fatigue. *Physiological Measurement*, *37*(11), 1993–2002. https://doi.org/10.1088/0967-3334/37/11/1993
20. Mangine, G. T., Hoffman, J. R., Wang, R., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Miramonti, A. A., La Monica, M. B., Fukuda, D. H., & Stout, J. R. (2016). Resistance training intensity and volume affect changes in rate of force development in resistance-trained men. *European Journal of Applied Physiology*, *116*(11), 2367–2374. https://doi.org/10.1007/s00421-016-3488-6
21. Jenkins, N. D. M., Housh, T. J., Miramonti, A. A., McKay, B. D., Yeo, N. M., Smith, C. M., Hill, E. C., Cochrane, K. C., & Cramer, J. T. (2016). Effects of rumenic acid rich conjugated linoleic acid supplementation on cognitive function and handgrip performance in older men and women. *Experimental Gerontology*, *84*, 1–11. https://doi.org/10.1016/j.exger.2016.08.004
22. Beyer, K. S., Fukuda, D. H., Miramonti, A. A., Church, D. D., Tanigawa, S., Stout, J. R., & Hoffman, J. R. (2016). Strength ratios are affected by years of experience in American collegiate rugby athletes: A preliminary study. *Isokinetics and Exercise Science*, *24*(3), 257–262. https://doi.org/10.3233/ies-160625
23. Wang, R., Hoffman, J. R., Tanigawa, S., Miramonti, A. A., La Monica, M. B., Beyer, K. S., Church, D. D., Fukuda, D. H., & Stout, J. R. (2016). Isometric mid-thigh pull correlates with strength, sprint and agility performance in collegiate rugby union players. *Journal of Strength & Conditioning Research*, *30*(11), 3051–3056. https://doi.org/10.1519/jsc.0000000000001416
24. La Monica, M. B., Fukuda, D. H., Miramonti, A. A., Beyer, K. S., Hoffman, M. W., Boone, C. H., Tanigawa, S., Wang, R., Church, D. D., Stout, J. R., & Hoffman, J. R. (2016). Physical differences between forwards and backs in American collegiate rugby players. *Journal of Strength & Conditioning Research*, *30*(9), 2382–2391. https://doi.org/10.1519/jsc.0000000000001388
25. La Monica, M. B., Fukuda, D. H., Beyer, K. S., Hoffman, M. W., Miramonti, A. A., Riffe, J. J., Baker, K. M., Fragala, M. S., Hoffman, J. R., & Stout, J. R. (2016). Altering work to rest ratios differentially influences fatigue indices during repeated sprint ability testing. *Journal of Strength & Conditioning Research*, *30*(2), 400–406. https://doi.org/10.1519/jsc.0000000000001122

26. Jenkins, N. D. M., Housh, T. J., Buckner, S. L., Bergstrom, H. C., Smith, C. M., Cochrane, K. C., Hill, E. C., Miramonti, A. A., Schmidt, R. J., Johnson, G. O., & Cramer, J. T. (2016). Four weeks of high- versus low-load resistance training to failure on the rate of torque development, electromechanical delay, and contractile twitch properties. *Journal of Musculoskeletal and Neuronal Interactions*, *16*(2), 135–144.
27. La Monica, M. B., Fukuda, D. H., Beyer, K. S., Hoffman, M. W., Miramonti, A. A., Fragala, M. S., Stout, J. R., & Hoffman, J. R. (2016). Evaluation of critical rest interval determined from repeated sprint ability testing. *Journal of Sports Medicine and Physical Fitness*, *56*(10), 1093–1102.
28. Wang, R., Fukuda, D. H., Stout, J. R., Robinson, E. H., Miramonti, A. A., Fragala, M. S., & Hoffman, J. R. (2015). Evaluation of electromyographic frequency domain changes during a three-minute maximal effort cycling test. *Journal of Sports Science & Medicine*, *14*(2), 452–458.
29. Mangine, G. T., Hoffman, J. R., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Miramonti, A. A., Wang, R., La Monica, M. B., Fukuda, D. H., Ratamess, N. A., & Stout, J. R. (2015). The effect of training volume and intensity on improvements in muscular strength and size in resistance-trained men. *Physiological Reports*, *3*(8), e12472. https://doi.org/10.14814/phy2.12472
30. Gonzalez, A. M., Hoffman, J. R., Wells, A. J., Mangine, G. T., Townsend, J. R., Jajtner, A. R., Wang, R., Miramonti, A. A., Pruna, G. J., La Monica, M. B., Bohner, J. D., Hoffman, M. W., Oliveira, L. P., Fukuda, D. H., Fragala, M. S., & Stout, J. R. (2015). Effects of time-release caffeine containing supplement on metabolic rate, glycerol concentration, and performance. *Journal of Sports Science & Medicine*, *14*, 322–332.
31. Robinson, E. H., Stout, J. R., Miramonti, A. A., Fukuda, D. H., Wang, R., Townsend, J. R., Mangine, G. T., Fragala, M. S., & Hoffman, J. R. (2014). High-intensity interval training and β-hydroxy-β-methylbutyric free acid improves aerobic power and metabolic thresholds. *Journal of the International Society of Sports Nutrition*, *11*, 16. https://doi.org/10.1186/1550-2783-11-16

Manuscripts in Progress

Completeness Thresholds Matter in Person-Mean Imputation and Prorated Scale Score Methods - Miramonti AA, Jeffries J, Riley JH, Bovaird JA — In progress

Evaluation of the feasibility of a two-method measurement design for the assessment of healthy physical activity behavior in youth - Miramonti AA, Bovaird JA, Krehbiel M, Franzen-Castle L — In progress

Electromyographic and mechanomyographic amplitude patterns of responses during isometric versus concentric dynamic constant external resistance leg extension muscle actions - Miramonti AA, Jenkins NDM, Gillen ZM — In progress

Improved critical power and muscle cross-sectional area following a progressive 4-week cycling interval training program in men and women - Fukuda DH, Stout JR, Robinson EH, Wang R, Mangine GT, Miramonti AA, Fragala MS, Hoffman JR — In progress

Validation of the HOME Inventory's Time Invariance in FFCWS: Insights into Home Environment Changes in Disadvantaged Parents Over a Decade - Yang Z, Miramonti AA, Choi JK, Jiang Q, Wang D — In progress

"It helps me deal with what's happening right now instead of letting my emotions get out of control." Exploring a Self-Compassion Group for Family Caregivers - Rasby S, Miramonti AA, Wilhite H — In progress

Racial and Gender Biases in Interpretations of Typical Infant/Toddler Behaviors - Albert R, Beeghly MJ, Brophy-Herb H, Gardner-Neblett N, Miramonti AA, Torquati J, Vallotton C — In progress

Technical & Evaluation Reports

Community Crops Classes 2025 Evaluation Report: Reach and Impacts of Classes Offered June 2023–August 2025 University of Nebraska Public Policy Center. - PytlikZillig LM, Miramonti AA — September 2025

Papers Presented at Professional Meetings

1. Wang, P., Miramonti, A. A., & Brown, S. (2026). From faculty DEI awareness to inclusive teaching: Insights from a multi-part survey study. *2026 American Educational Research Association Annual Meeting*.
2. George, R. M., Dev, D. A., Miramonti, A. A., Yan, C., & Hughes, M. (2024, May). Examining disparities in the association of obesity with sociodemographic factors along the rural-urban continuum. *2024 International Society for Behavioral Nutrition and Physical Activity Conference*.
3. George, R. M., Dev, D. A., Hasnin, S., Hillburn, C., Smith, J., Sisson, S. B., Tovar, A., & Miramonti, A. A. (2024, May). Family style meal service is associated with reduced plate waste in Nebraska family child care homes. *2024 International Society for Behavioral Nutrition and Physical Activity Conference*.
4. Miramonti, A. A., Jeffries, J., Riley, J. H., & Bovaird, J. A. (2024). Completeness thresholds matter in person-mean imputation and prorated scale score methods. *2024 American Psychological Association Annual Convention*.
5. Jeffries, J., Miramonti, A. A., Riley, J. H., & Bovaird, J. A. (2024). Untangling single imputation methods: A focus on person-mean imputation and prorated scale scores. *2024 American Psychological Association Annual Convention*.
6. Harper, E. E., Miramonti, A. A., Conners, F. A., & Loveall, S. J. (2024). Impact of sensory differences on executive function in school-age individuals with Down syndrome. *2024 Gatlinburg Conference on Research and Theory in Intellectual and Developmental Disabilities*.
7. Jeffries, J., & Miramonti, A. A. (2024). When construct validity meets network analysis: An exploratory graph analysis simulation study. *2024 American Educational Research Association Annual Meeting, Division d — Measurement and Research Methodology*.
8. Kim, S., Kantamneni, N., Ahmad Ridzuan, N. H., & Miramonti, A. A. (2023). Work-family balance and well-being for women of color with children. *2023 American Psychological Association Annual Convention*.
9. Clark, C. A. C., Miramonti, A. A., & Chen, D. (2021). Segregation of task-positive and negative functional neural networks uniquely relates to children's executive control in middle childhood. *2021 FLUX Virtual Congress*.
10. Miramonti, A. A., Bovaird, J. A., Franzen-Castle, L., & Krehbiel, M. (2019, May). Unidimensionality and internal consistency reliability of step counts and floors climbed in 4th–5th grade students. *2019 American College of Sports Medicine Annual Meeting*.
11. Miramonti, A. A., Bovaird, J. A., Franzen-Castle, L., & Krehbiel, M. (2019). Measurement invariance of a latent model of child physical activity based on child and parent surveys and wearable physical activity tracker data. *2019 National Strength & Conditioning Association National Conference*.
12. Miramonti, A. A., Bovaird, J. A., Franzen-Castle, L., & Krehbiel, M. (2019). Changes in self-reported healthy eating behavior frequency and nutrition knowledge among youth participating in the WeCook program. *Nutrition 2019 Conference of the American Society of Nutrition*.
13. Miramonti, A. A., Krehbiel, M., Bovaird, J. A., Dunker, T., & Franzen-Castle, L. (2019). A program evaluation of an afterschool nutrition, cooking, and physical activity program: WeCook: Food with fun and fitness. *2019 Nebraska Healthy Kids Summit Research Fair*.
14. Tomko, P. M., Miramonti, A. A., Hill, E. C., Smith, C. M., Cochrane-Snyman, K. C., Colquhoun, R. J., Housh, T. J., Cramer, J. T., & Jenkins, N. D. M. (2018, May). Mechanomyographic amplitude is sensitive to neuromuscular adaptations following high- versus low-load resistance training. *2018 American College of Sports Medicine Annual Meeting*.
15. Dunker, T., Miramonti, A. A., & Krehbiel, M. (2018). Assessing nutrition knowledge using a healthy plate photo method among youth participating in the WeCook program. *2018 National Association of Extension 4-h Agents Conference*.
16. Miramonti, A. A., Jenkins, N. D. M., Hill, E. C., Smith, C. M., Housh, T. J., & Bovaird, J. A. (2018). Electromyographic and mechanomyographic amplitude response patterns during isometric vs. Concentric dynamic constant external resistance leg extension muscle actions. *2018 National Strength & Conditioning Association National Conference*.
17. Miramonti, A. A., Cramer, J. T., Jenkins, N. D. M., Gillen, Z. M., McKay, B. D., & Leutzinger, T. J. (2017, May). Effects of speed and agility training on combine performance in young male athletes. *2017 American College of Sports Medicine Annual Meeting*.
18. Gillen, Z. M., Miramonti, A. A., McKay, B. D., Jenkins, N. D. M., Leutzinger, T. J., & Cramer, J. T. (2017, May). Power push-up tests performed from the knees and toes in young male athletes. *2017 American College of Sports Medicine Annual Meeting*.

19. McKay, B. D., Jenkins, N. D. M., Miramonti, A. A., Leutzinger, T. J., & Cramer, J. T. (2017, May). Test-retest reliability of the 40-yard dash and vertical jump assessments in youth athletes. *2017 American College of Sports Medicine Annual Meeting*.
20. Church, D. D., Hoffman, J. R., Mangine, G. T., Jajtner, A. R., Townsend, J. R., Gonzalez, A. M., Beyer, K. S., Wang, R., Boone, C. H., La Monica, M. B., Miramonti, A. A., Wells, A. J., Fukuda, D. H., & Stout, J. R. (2017, May). The response of leukemia inhibitory factor to high intensity and high volume resistance training in trained men. *2017 American College of Sports Medicine Annual Meeting*.
21. Miramonti, A. A., Fischer, J. A., Sehi, N., Veseth, N., & Cramer, J. T. (2017). Eight lessons of CATCH® curricula for Hispanic/Latino and non-Hispanic/Latino students in Nebraska. *2017 Food & Nutrition Conference & Expo*.
22. Miramonti, A. A., Gillen, Z. M., McKay, B. D., Leutzinger, T. J., Mendez, A. I., & Cramer, J. T. (2017). Competitive youth soccer participation improves sprint and broad jump performance. *2017 National Strength & Conditioning Association National Conference*.
23. Gillen, Z. M., Miramonti, A. A., McKay, B. D., Leutzinger, T. J., & Cramer, J. T. (2017). Age-related differences in agility drill performances among youth athletes: Influences of growth and development. *2017 National Strength & Conditioning Association National Conference*.
24. McKay, B. D., Herda, A. A., Stout, J. R., Miramonti, A. A., Gillen, Z. M., & Cramer, J. T. (2017). Specific adaptations in older men and women performing exercise with self-selected intensities. *2017 National Strength & Conditioning Association National Conference*.
25. Miramonti, A. A., Housh, T. J., Jenkins, N. D. M., Cochrane, K. C., Hill, E. C., Smith, C. M., Schmidt, R. J., Johnson, G. O., & Cramer, J. T. (2016, May). Relationships among cognitive function and handgrip strength and endurance in older men and women. *2016 American College of Sports Medicine Annual Meeting*.
26. Jenkins, N. D. M., Miramonti, A. A., Hill, E. C., Smith, C. M., Cochrane, K. C., Housh, T. J., & Cramer, J. T. (2016, May). Are voluntary activation determined by the interpolated twitch technique and mechanomyographic amplitude synonymous? *2016 American College of Sports Medicine Annual Meeting*.
27. Hill, E. C., Housh, T. J., Smith, C. M., Cochrane, K. C., Jenkins, N. D. M., Miramonti, A. A., Cramer, J. T., Schmidt, R. J., & Johnson, G. O. (2016, May). Effects of work-to-rest ratios on peak torque and neuromuscular responses during submaximal, isometric muscle actions. *2016 American College of Sports Medicine Annual Meeting*.
28. Cochrane, K. C., Housh, T. J., Smith, C. M., Hill, E. C., Jenkins, N. D. M., Miramonti, A. A., Schmidt, R. J., Johnson, G. O., & Cramer, J. T. (2016, May). Physiological responses underlying the perception of effort during moderate and heavy intensity cycle ergometry. *2016 American College of Sports Medicine Annual Meeting*.
29. Smith, C. M., Housh, T. J., Hill, E. C., Cochrane, K. C., Jenkins, N. D. M., Miramonti, A. A., Cramer, J. T., Schmidt, R. J., & Johnson, G. O. (2016, May). Effects of varied intensity on torque and neuromuscular parameters during intermittent isometric muscle actions. *2016 American College of Sports Medicine Annual Meeting*.
30. Riffe, J. J., Stout, J. R., Robinson, E. H., Miramonti, A. A., Beyer, K. S., Wang, R., La Monica, M. B., Church, D. D., Muddle, T. W. D., & Hoffman, J. R. (2016). Methodological comparison of PWCFT estimation in response to high-intensity interval training. *2016 American College of Sports Medicine Annual Meeting*.
31. Cramer, J. T., McKay, B. D., Jenkins, N. D. M., Miramonti, A. A., & Weinstein, M. (2016). Normative reference values for high school American football combine test results: Effects of allometric scaling to normalize for body mass. *2016 National Strength & Conditioning Association National Conference*.
32. Hill, E. C., Housh, T. J., Smith, C. M., Cochrane, K. C., Jenkins, N. D. M., Miramonti, A. A., Cramer, J. T., Schmidt, R. J., & Johnson, G. O. (2016). Gender-related differences in muscle fatigue. *2016 National Strength & Conditioning Association National Conference*.
33. Jenkins, N. D. M., Housh, T. J., Miramonti, A. A., Smith, C. M., Hill, E. C., Cochrane, K. C., & Cramer, J. T. (2016). Effects of rumenic acid rich conjugated linoleic acid supplementation on handgrip performance and cognitive function in older men. *2016 National Strength & Conditioning Association National Conference*.
34. Smith, C. M., Housh, T. J., Hill, E. C., Cochrane, K. C., Jenkins, N. D. M., Miramonti, A. A., Schmidt, R. J., Johnson, G. O., & Cramer, J. T. (2016). Neuromuscular responses during fatiguing intermittent isometric muscle actions. *2016 National Strength & Conditioning Association National Conference*.
35. Mangine, G. T., Hoffman, J. R., Gamazo, T., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Wang, R., Miramonti, A. A., La Monica, M. B., Fukuda, D. H., Ratamess, N. A., & Stout, J. R. (2016). Influence of competitive level and position in American rugby on body composition and measures of performance. *2016 National Strength & Conditioning Association National Conference*.
36. La Monica, M. B., Fukuda, D. H., Beyer, K. S., Hoffman, M. W., Miramonti, A. A., Riffe, J. J., Baker, K. M., Fragala, M. S., Hoffman, J. R., & Stout, J. R. (2015, May). Alteration of work to rest ratios influences repeated sprint ability performance. *2015 American College of Sports Medicine Annual Meeting*.
37. Church, D. D., Mangine, G. T., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Miramonti, A. A., Wang, R., La Monica, M. B., Fukuda, D. H., Stout, J. R., & Hoffman, J. R. (2015, May). High-intensity resistance training increases upper body muscle and bone variables in resistance-trained males. *2015 American College of Sports Medicine Annual Meeting*.

38. Robinson, E. H., Stout, J. R., Fukuda, D. H., Wang, R., Miramonti, A. A., Fragala, M. S., Mangine, G. T., Townsend, J. R., & Hoffman, J. R. (2015, May). HMB supplementation and high-intensity interval training improves efficiency of muscle recruitment more than training alone. *2015 American College of Sports Medicine Annual Meeting*.
39. Miramonti, A. A., Fukuda, D. H., Stout, J. R., & Hoffman, J. R. (2015, May). Comparison of collegiate track divisions using the distance-time relationship. *2015 American College of Sports Medicine Annual Meeting*.
40. Fukuda, D. H., Stout, J. R., Robinson, E. H., Wang, R., Miramonti, A. A., Fragala, M. S., & Hoffman, J. R. (2015, May). Relationship between critical power and heart rate deflection point as estimates of heavy-severe exercise intensities. *2015 American College of Sports Medicine Annual Meeting*.
41. Miramonti, A. A., Stout, J. R., Robinson, E. H., Wang, R., Fukuda, D. H., & Hoffman, J. R. (2015). The effects of high-intensity interval training and β-hydroxy-β-methylbutyric free acid supplementation on the electromyographic fatigue threshold (EMGFT). *2015 National Strength & Conditioning Association National Conference*.
42. Beyer, K. S., Fukuda, D. H., Miramonti, A. A., Robinson, E. H., Stout, J. R., & Hoffman, J. R. (2015). Hamstrings:quadriceps ratio is affected by years of experience in American collegiate rugby athletes independent of age and body weight. *2015 National Strength & Conditioning Association National Conference*.
43. Fukuda, D. H., Miramonti, A. A., Hoffman, M. W., Wang, R., Beyer, K. S., Riffe, J. J., Tanigawa, S., Stout, J. R., & Hoffman, J. R. (2015). Association between spatial awareness and medium intensity cruising/striding during a collegiate rugby union match. *2015 National Strength & Conditioning Association National Conference*.
44. Hoffman, M. W., Stout, J. R., Miramonti, A. A., La Monica, M. B., Wang, R., Beyer, K. S., Tanigawa, S., Fukuda, D. H., & Hoffman, J. R. (2015). A thirty-second all-out test using a non-motorized treadmill correlates to power and quickness in collegiate rugby players. *2015 National Strength & Conditioning Association National Conference*.
45. Mangine, G. T., Hoffman, J. R., Gonzalez, A. M., Wells, A. J., Jajtner, A. R., Beyer, K. S., Boone, C. H., Wang, R., Miramonti, A. A., La Monica, M. B., Fukuda, D. H., Witta, E. E., Ratamess, N. A., & Stout, J. R. (2015). The influence of exercise-induced hormonal elevation on muscle growth in resistance-trained men. *2015 National Strength & Conditioning Association National Conference*.
46. La Monica, M. B., Fukuda, D. H., Miramonti, A. A., Beyer, K. S., Hoffman, M. W., Tanigawa, S., Wang, R., Stout, J. R., & Hoffman, J. R. (2015). Performance profiles of championship-level collegiate rugby players: Comparison between athletes of different position groups. *2015 National Strength & Conditioning Association National Conference*.
47. Riffe, J. J., Fukuda, D. H., Robinson, E. H., Miramonti, A. A., La Monica, M. B., Hoffman, J. R., & Stout, J. R. (2015). The reliability of a D-Max method for estimating electromyography fatigue threshold during cycle ergometry. *2015 National Strength & Conditioning Association National Conference*.
48. Tanigawa, S., Wang, R., Miramonti, A. A., Hoffman, M. W., Fukuda, D. H., Stout, J. R., & Hoffman, J. R. (2015). Evaluation of performance measures between field tests and GPS traction in collegiate rugby players. *2015 National Strength & Conditioning Association National Conference*.
49. Varanoske, A. N., Fukuda, D. H., Boone, C. H., Wang, R., Miramonti, A. A., La Monica, M. B., Hoffman, M. W., Riffe, J. J., Tanigawa, S., Stout, J. R., & Hoffman, J. R. (2015). Evaluation of bilateral differences in lower body power production in collegiate rugby players. *2015 National Strength & Conditioning Association National Conference*.
50. Wang, R., Hoffman, J. R., Fukuda, D. H., Tanigawa, S., Miramonti, A. A., & Stout, J. R. (2015). Isometric mid-thigh pull is correlated with agility and short sprint performance in collegiate rugby players. *2015 National Strength & Conditioning Association National Conference*.
51. Miramonti, A. A., Fukuda, D. H., Stout, J. R., Robinson, E. H., Wang, R., Mangine, G. T., Townsend, J. R., Bohner, J. D., Fragala, M. S., & Hoffman, J. R. (2014, May). Gender differences in anaerobic working capacity, but not critical power, during an all-out cycling test. *2014 American College of Sports Medicine Annual Meeting*.
52. Robinson, E. H., Fukuda, D. H., Stout, J. R., Wang, R., Miramonti, A. A., Townsend, J. R., Mangine, G. T., Jajtner, A. R., Wells, A. J., Gonzalez, A. M., McCormack, W. P., Bohner, J. D., La Monica, M. B., Reardon, D., Oliveira, L. P., Fragala, M. S., & Hoffman, J. R. (2014, May). Gender-specific differences in the relationship between muscle morphology and neuromuscular economy. *2014 American College of Sports Medicine Annual Meeting*.
53. Wang, R., Fukuda, D. H., Stout, J. R., Robinson, E. H., Miramonti, A. A., Townsend, J. R., Mangine, G. T., Jajtner, A. R., Wells, A. J., Gonzalez, A. M., McCormack, W. P., Bohner, J. D., Fragala, M. S., & Hoffman, J. R. (2014, May). Evaluation of EMG frequency domain changes during a three-minute maximal effort cycling test. *2014 American College of Sports Medicine Annual Meeting*.
54. Fukuda, D. H., Stout, J. R., Robinson, E. H., Wang, R., Mangine, G. T., Miramonti, A. A., Fragala, M. S., & Hoffman, J. R. (2014). Improved critical power and muscle cross-sectional area following a progressive 4-week cycling interval training program in men and women. *2014 National Strength & Conditioning Association National Conference*.

55. Gonzalez, A. M., Hoffman, J. R., Wells, A. J., Mangine, G. T., Townsend, J. R., Jajtner, A. R., Wang, R., Miramonti, A. A., Pruna, G. J., La Monica, M. B., Bohner, J. D., Hoffman, M. W., Oliveira, L. P., Fukuda, D. H., Fragala, M. S., & Stout, J. R. (2014). Pharmacokinetics of caffeine administered in a time-release versus regular tablet form. *Journal of the International Society of Sports Nutrition*, *11*(Suppl 1), P23.
56. Wells, A. J., Hoffman, J. R., Gonzalez, A. M., Beyer, K. S., Jajtner, A. R., Townsend, J. R., La Monica, M. B., Miramonti, A. A., Hoffman, M. W., Oliveira, L. P., Fukuda, D. H., Fragala, M. S., & Stout, J. R. (2014). Examination of the health and safety aspects of 28-days ingestion of a supplement containing slow-release caffeine. *Journal of the International Society of Sports Nutrition*, *11*(Suppl 1), P17.

Mentoring Experience

Graduate Research Assistant Supervisor - Nebraska Evaluation and Research Center, University of Nebraska–Lincoln — 2024–present

Instructor, non-credit methods sessions - Informal graduate methods training for six students; precursor to EDPS 845, University of Nebraska–Lincoln — Fall 2025

McNair Graduate Mentor - Ronald E. McNair Scholars Program, University of Nebraska–Lincoln — 2016–2017

UCARE Graduate Mentor - Undergraduate Creative Activities and Research Experience (UCARE) Program, University of Nebraska–Lincoln — 2017

Additional Professional Training

Nebraska Academy for Methodology, Analytics, & Psychometrics: Nebraska Methodology Applications

Regression Discontinuity Designs in Social Science Research: Causal Inference of Cutoff-Based Programs - HyeonJin Yoon — March 2020

Pursuing Causal Inferences in the Absence (or Failure) of Random Assignment: An Introduction to Propensity Score Analysis - Natalie Koziol — February 2020

Improving Data Quality Through Qualtrics Survey Design - Jared Stevens — February 2019

Addressing One Research Question Using Multiple Methodological Approaches - Marc Goodrich — April 2019

Time is on my side: What British rock bands can teach us about designing longitudinal studies - Matt Fritz — February 2018

An Introduction to Machine Learning - Stephen Scott — November 2017

Nebraska Academy for Methodology, Analytics, & Psychometrics: Emerging Scholars Series

Evidence-Based Assessment: From simple clinical judgments to statistical learning - Eric Youngstrom — April 2018

Does Model Fit have a Validity Problem? - Mike Edwards — September 2017

Training and Assessing Communication Skills Using Virtual Human Technology - Tim Guetterman — April 2019

Professional Service

Conference Abstract Reviewer - National Strength & Conditioning Association — 2016, 2019

Reviewer - Journal of Strength and Conditioning Research — 2016–2020

Reviewer - Research Quarterly for Exercise and Sport — 2019–2020

Reviewer - Women in Sport and Physical Activity Journal — 2026–present

Event Volunteer - Nebraska Women in Tech Summit, AKSARBEN Foundation — 2026

Graduate Student Volunteer Poster Judge - UNL CEHS Undergraduate Research Fair — 2019

Professional Associations

American Psychological Association - Member — 2024–2025

American Statistical Association - Member — 2019–2020

Society for Transparency, Openness, and Replication in Kinesiology - Member — 2019–present

National Strength & Conditioning Association - Member, 2014–2020; Research Consortium Member, 2019–2020

American College of Sports Medicine - Member — 2013–2020

National Intramural Recreational Sports Association - Member — 2009–2010

Professional Credentials

Certified Strength & Conditioning Specialist - National Strength & Conditioning Association — Expired

Adult CPR/AED, Child CPR and First Aid Certification - American Red Cross — Expired

Awards, Fellowships, & Scholarships

Plake Fellowship - Department of Educational Psychology, University of Nebraska–Lincoln — 2024

AAUW Fellowship - American Association of University Women, Lincoln, NE Chapter — 2019–2020

Women's Scholarship - National Strength & Conditioning Association — 2016, 2017

Othmer Fellowship - University of Nebraska–Lincoln — 2015–2018

Graduate Research Excellence Fellowship - University of Central Florida — 2014–2015

Pegasus Scholarship - University of Central Florida — 2007–2010

Finalist, National Merit Scholarship Competition - — 2007

Software

Nebraska Suicide Mortality Review Reporting Application - Lead developer, R / Shiny — 2026–present

- Application supporting Nebraska's veteran suicide mortality review process: integrates and reconciles successive Nebraska Violent Death Reporting System (NEVDRS) exports, detects and tracks record-level changes between exports, supports structured case annotation, and generates both individual case reports and the annual report to the Legislature. Internal; not publicly released.

NSWERS Data Explorer & Pipeline Visualization - Author, R / Shiny — 2025–present

- Internal application for the Nebraska Statewide Workforce and Educational Reporting System data warehouse: locates which tables hold which variables, retrieves definitions, and traces data lineage back to the queries and source tables each variable depends on. Improves pipeline transparency, makes problems easier to pinpoint, and shortens onboarding for the team. Not publicly released.

Recode Studio - Author & maintainer, R / Shiny — Ongoing

- Dataset-agnostic R/Shiny app for cleaning messy string variables without hand-writing R: browse values, flag likely typos via spellcheck, build and preview recode rules, and export both a reusable rule set and runnable R code. Open source: github.com/amelia-m/recode-studio

Table Relationship Explorer - Author & maintainer, R / Shiny — Ongoing

- Interactive tool for discovering and visualizing primary/foreign-key relationships across tabular data: multi-signal key inference with confidence scoring, interactive entity-relationship diagrams, and exportable reports across many file and database formats (with a Python/Streamlit port). Open source: github.com/amelia-m/table-explorer

Technical Skills

Laboratory Assessments

- Heart rate (Polar monitors), blood pressure, anthropometrics
- Resting metabolic rate and maximal oxygen uptake (ParvoMedics TrueOne 2400, Sensormedics Vmax Encore); Wingate and 3-minute critical power testing (Lode Corrival cycle ergometer)
- Isometric, isokinetic, and isotonic strength testing (Biodex System)
- Electromyographic and mechanomyographic data collection and processing (Noraxon Desktop DTS, Biopac)
- Sport performance testing: sprints, agility, jumps, and force plate assessments using custom LabVIEW data collection interfaces
- Physical activity monitoring (Actigraph wGT3X-BT)
- Finger prick blood sample collection, basic sample processing

Software & Computing

Statistical & Psychometric - R Statistical Software (tidyverse; lavaan and related psychometric packages; MplusAutomation and semPlot for SEM workflows); Mplus; SAS; SPSS (including PROCESS macro); MATLAB; Jamovi; JASP

Synthetic Data Generation & Evaluation - tidysynthesis; syntheval (R packages, Urban Institute)

Data Collection - Qualtrics; REDCap; LabVIEW (data collection, signal processing, and analysis); Biopac (EMG/MMG data collection)

Data Management & Databases - dbplyr (R); SQL; Microsoft Access

Visualization & Interactive Applications - ggplot2 (R); gt (R); Shiny (R); Veusz

Reproducible Reporting & Publishing - Quarto (course websites, slide decks, parameterized reports, this CV); R Markdown / knitr; Markdown; LaTeX / TinyTeX

Research Workflow & Reference Management - targets (R package for reproducible pipelines); renv; Zotero; Obsidian; Open Science Framework (OSF)

Programming & Version Control - Git (GitHub, GitLab); Python (basic & LLM-assisted); Unix / Bash

Development Environments & Editors - Positron; RStudio; Visual Studio Code; Notepad++

LLM-Assisted Development - Claude Code; Gemini / Gemini CLI; GitHub Copilot; Antigravity IDE; Cursor

Productivity & Collaboration - Microsoft Office Suite (Excel, PowerPoint, Word, Teams, SharePoint); Adobe Acrobat Pro DC; Adobe Illustrator

Online Profiles

Google Scholar - https://scholar.google.com/citations?hl=en&user=AcbJlasAAAAJ

LinkedIn - https://www.linkedin.com/in/amelia-miramonti/

ResearchGate - https://www.researchgate.net/profile/Amelia-Miramonti

ORCiD - https://orcid.org/0000-0002-0277-4064

GitHub - https://github.com/amelia-m

GitLab (UNL) - https://git.unl.edu/amiramonti

OSF - https://osf.io/zk8rv/

Lens - https://www.lens.org/lens/profile/528341700/scholar

Scopus - https://www.scopus.com/authid/detail.uri?authorId=56123562400